

1. Calculation of Architectural Metrics

1.1 Single Point Fault Metric (SPFM)

$$SPFM = 1 - \frac{\sum (\lambda_{SP} + \lambda_{RF})}{\sum \lambda}$$
$$SPFM = 1 - (\frac{0.87}{115.358})$$

SPFM = acc. ISO 26262-5, table 4

1.2 Latent Fault Metric (LFM)

$$LFM = 1 - \frac{\sum \lambda_{LF}}{\sum \lambda - (\lambda_{SP} + \lambda_{RF})}$$
$$LFM = 1 - (\frac{20.29}{115.229})$$

LFM = acc. ISO 26262-5, table 5

2a. Calculation of the Probabilistic Metric (PMHF)

acc. to ISO 26262-5 (Annex F.2) we can estimate ☐

$$PMHF_{est} = \sum \lambda_{SP} + \sum \lambda_{RF} + \sum \lambda_{MPF det} * \sum \lambda_{LF} * T_{lifetime}$$
$$PMHF_{est} = 0.74 + 0.13 + 29.01 * 20.29 * 5475 \text{ h}$$

PMHF ≈ acc. ISO 26262-5, table 6

2b. Alternative: Check of "Failure Rate Classes"

Check of requirements according
ISO 26262-5, table 8,9,10

Result (FRC) acc. ISO 26262-5, table 8,9,10